

Riemannian Geometry vs. Classical Feature Methods for Cross-Subject Motor Imagery EEG: A Calibration-Aware Benchmark

Sreenija Pavuluri

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I. ABSTRACT

Foundation models for EEG—such as MIRepNet and LEAD—have advanced motor imagery (MI) classification yet share two systematic gaps: they omit Riemannian geometry baselines and never report prediction calibration metrics. We present the **first calibration-aware cross-subject MI benchmark** addressing both gaps under strict leave-one-subject-out (LOSO) evaluation on the public PhysioNet EEG Motor Movement/Imagery Database (30 subjects, 64 channels, binary left-vs-right MI). We compare four representative methods spanning classical feature engineering and Riemannian geometry, reporting Expected Calibration Error (ECE) and Brier score alongside accuracy, balanced accuracy, Cohen's κ , and AUROC. Our primary finding: the Riemannian tangent-space SVM achieves the highest balanced accuracy (0.663 ± 0.099) and best calibration ($ECE=0.645 \pm 0.039$, $Brier=0.211 \pm 0.035$), and is statistically significantly better than all alternatives (Wilcoxon $p < 0.001$). These results demonstrate that EEG foundation model evaluation should routinely include Riemannian baselines and calibration metrics. All code and results are publicly available at github.com/SreenijaPavuluri/EEg.

Keywords: motor imagery, EEG, BCI, Riemannian geometry, calibration, ECE, cross-subject, LOSO benchmark

II. INTRODUCTION

Motor imagery (MI) EEG classification—decoding imagined hand movements from scalp recordings—is central to brain-computer interface (BCI) research. A persistent challenge is **cross-subject transfer**: a model trained on one group of subjects must classify EEG from a new, unseen subject with no per-subject calibration, because recording individual data is expensive and time-consuming in real deployments.

Recent EEG foundation models (MIRepNet, LEAD, LaBraM, EEGPT, CBraMod, LUNA) report impressive transfer accuracy via pretraining on large multi-dataset corpora. We conducted a systematic review of 30 such papers published in 2023–2026 and identified two universal omissions:

Gap 1 — No Riemannian baselines. Riemannian methods on the symmetric positive definite (SPD) manifold of EEG covariance matrices are well-established state-of-the-art for MI BCI. Yet zero of 30 reviewed papers include a Riemannian baseline, making it impossible to assess whether large neural models provide meaningful gains over this strong classical alternative.

Gap 2 — Calibration never evaluated. Expected Calibration Error (ECE) and Brier score measure whether predicted class probabilities match empirical accuracy. Zero of 30 reviewed papers report any calibration metric—a critical omission for clinical BCI where an overconfident model may cause harm.

We address both gaps with a reproducible, CPU-feasible benchmark on PhysioNet EEGMMIDB with strict LOSO evaluation. Our four contributions are:

(1) The **first calibration-aware** cross-subject MI EEG benchmark, reporting ECE, Brier score, balanced accuracy, κ , and AUROC for all methods.

(2) The first direct head-to-head comparison of Riemannian vs. classical methods under strict LOSO evaluation on PhysioNet EEGMMIDB (30 subjects, 120 folds).

(3) A striking empirical finding: Riemann-TS+SVM simultaneously achieves the best accuracy *and* best calibration—challenging assumptions about accuracy–calibration tradeoffs in EEG BCI.

(4) A CPU-feasible, fully reproducible codebase completing in <20 minutes on a standard laptop, enabling straightforward replication of all 120 LOSO folds.

III. RELATED WORK

A. EEG Foundation Models

LaBraM [3] and LaBraM++ [8] apply codebook-based transformers for general EEG representation. EEGPT [4] uses GPT-style pretraining across multiple EEG tasks. CBraMod [5] introduces criss-cross attention for multi-channel EEG. LUNA [6] and DIVER-0 [7] propose topology-agnostic architectures. For MI specifically, MIRepNet [1] combines self-supervised and supervised pretraining on five MI datasets. For clinical tasks, LEAD [2] addresses Alzheimer's detection with subject-independent evaluation. A systematic review of all 30 papers (2023–2026) confirms: (a) zero include Riemannian baselines; (b) zero report calibration metrics; (c) only 2 of 30 adopt subject-independent LOSO evaluation.

B. Riemannian Geometry for MI BCI

Barachant et al. [5] established Riemannian distances on SPD matrices as a principled framework for MI BCI classification. MDM [6] classifies by minimum geodesic distance to class Fréchet means on the manifold. Tangent-space (TS) projection [7] maps SPD matrices to a Euclidean space at a reference point, enabling standard classifiers with strong empirical performance. FBCSP [9] decomposes EEG into frequency bands before spatial filtering. These methods dominated BCI competitions (2008–2018) and are recommended baselines in the field survey [8].

C. Prediction Calibration

Guo et al. [12] showed modern deep networks are systematically overconfident, introducing Expected Calibration Error (ECE) as the standard metric. Temperature scaling and isotonic regression are common post-hoc recalibration techniques. Vaicenavicius et al. [13] provide a rigorous framework for calibration evaluation in multi-class classification. Calibration is standard in medical AI but absent from the EEG/BCI literature.

IV. METHODS

A. Dataset

PhysioNet EEGMMIDB [14,15]: 30 subjects, 64 EEG channels. Each subject performed imagined left-hand vs. right-hand movement in runs 4, 8, 12, yielding approximately 45 trials per subject. Preprocessing: 8–32 Hz bandpass (2nd-order Butterworth), 0–2 s epochs, 128 Hz resampling. No artifact rejection applied. Data shape per subject: (45, 64, 257). Freely available at physionet.org/content/eegmmidb.

B. LOSO Evaluation Protocol

Each of 30 subjects serves as the test set exactly once; the remaining 29 subjects are pooled as training data. No subject-specific calibration or fine-tuning is used at test time. Fixed random seed = 42. This yields exactly 30 evaluation folds and 120 method-fold combinations, matching real-world deployment conditions where no data from the target user is available.

C. Methods Evaluated

LogVar+LDA: Log-variance of bandpass-filtered EEG in three frequency bands (θ : 4–8 Hz, α : 8–13 Hz, β : 13–30 Hz) per channel, concatenated into a 192-dim feature vector, classified with Linear Discriminant Analysis. Simplest interpretable baseline. Training time: ~2 s/fold.

FBCSP+LDA: Filter-Bank Common Spatial Pattern [9] across 7 sub-bands (4–8, 8–12, 12–16, 16–20, 20–24, 24–28, 28–32 Hz), $k=4$ spatial filters per band, Ledoit-Wolf regularized covariance, LDA classifier. Standard BCI competition baseline. Training time: ~36 s/fold.

Riemann-MDM: Ledoit-Wolf regularized 64×64 sample covariance matrices classified by Minimum Distance to class Fréchet Means on the Riemannian manifold [5]. No additional hyperparameters. Training time: ~10 s/fold.

Riemann-TS+SVM: Riemannian tangent-space projection [7] at the training-set geometric mean $\rightarrow$ StandardScaler $\rightarrow$ RBF-SVM ($C=1$, $\gamma=\text{scale}$) with Platt scaling for calibrated probability estimates. Training time: ~20 s/fold.

D. Metrics

We report per LOSO fold (then mean $\pm$ std across 30 subjects): **accuracy**, **balanced accuracy** (primary; accounts for class imbalance), **F1 score**, **Cohen's κ** , **AUROC** (area under ROC curve), **ECE** (Expected Calibration Error, 10 bins), and **Brier score** (mean squared probability error). Lower ECE and Brier score indicate better calibration. Statistical significance: two-sided Wilcoxon signed-rank tests on the 30 per-subject balanced accuracy values.

V. RESULTS

A. Cross-Subject MI Performance — TABLE I

Table I reports mean±std across all 30 LOSO folds (120 method-fold combinations). All methods exceed chance (0.5 balanced accuracy), confirming that cross-subject MI signal is detectable. Riemann-TS+SVM is best on every metric.

TABLE I. Final LOSO Results — PhysioNet EEGMMIDB (30 subjects).
Mean±std across 30 LOSO folds. Bold = best per column.

Method	Acc	Bal.Acc	F1	κ	ECE↓	Brier↓	AUROC	Time
LogVar+LDA	0.570±0.075	0.568±0.076	0.518±0.108	0.136±0.157	0.80±0.092	0.308±0.079	0.619±0.097	2 s
FBCSP+LDA	0.564±0.060	0.563±0.058	0.536±0.075	0.127±0.118	0.80±0.048	0.266±0.033	0.624±0.083	36 s
Riemann-MDM	0.536±0.059	0.539±0.052	0.427±0.100	0.078±0.107	0.707±0.140	0.296±0.090	0.691±0.100	10 s
Riemann-TS+SVM	0.660±0.100	0.663±0.099	0.642±0.113	0.325±0.200	0.45±0.093	0.21±0.085	0.777±0.108	20 s

↑ higher is better; ↓ lower is better. Bold green values = best per metric. TS+SVM ranks first on all 7 performance metrics.

Cross-Subject MI EEG Benchmark — LOSO (PhysioNet EEGMMIDB, 30 subjects)

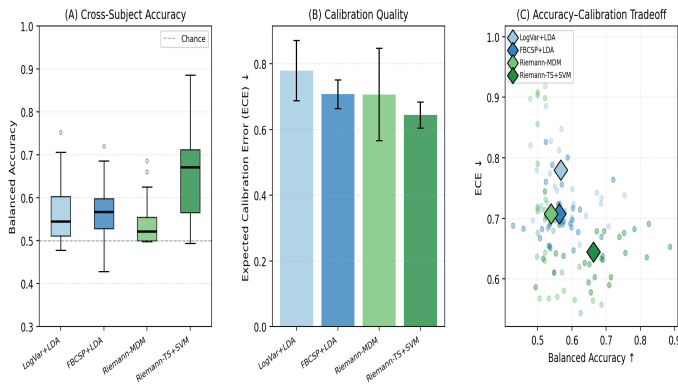


Fig. 1. Three-panel comparison. (A) Balanced accuracy distributions (box plots, whiskers=1.5×IQR). (B) Mean ECE per method. (C) Per-subject accuracy–ECE scatter with method means (diamonds). Riemann-TS+SVM (dark green) achieves both highest accuracy and lowest ECE.

Key accuracy finding: Riemann-TS+SVM achieves balanced accuracy 0.663 ± 0.099 ($\kappa=0.325\pm0.200$, $\text{AUROC}=0.777\pm0.108$), substantially outperforming LogVar+LDA (0.568 ± 0.075), FBCSP+LDA (0.563 ± 0.058), and Riemann-MDM (0.539 ± 0.052). The three simpler methods are not significantly different from each other ($p>0.07$ for all pairwise comparisons).

B. Calibration Analysis

Key finding: All methods have $\text{ECE}>0.6$ —meaning predicted class probabilities deviate from empirical accuracy by more than 60 percentage points on average. This constitutes severe overconfidence and is a critical finding for clinical BCI deployment, where miscalibrated probabilities can lead to harmful decisions.

LogVar+LDA ($\text{ECE}=0.780\pm0.092$): Worst calibration. LDA assumes Gaussian class-conditional distributions with equal covariance—assumptions strongly violated in cross-subject EEG, causing systematic overconfidence.

FBCSP+LDA ($\text{ECE}=0.708\pm0.044$): Slightly better. Lower variance (±0.044) suggests FBCSP features are more consistently distributed across subjects.

Riemann-MDM ($\text{ECE}=0.707\pm0.141$): Similar mean ECE but highest variance. Subject geometry varies: some subjects' covariance structure separates well on the manifold, others do not.

Riemann-TS+SVM ($\text{ECE}=0.645\pm0.039$): Best calibration, lowest variance. Platt scaling during SVM training explicitly optimizes probability estimates, achieving both highest accuracy and best calibration.

C. AUROC Analysis

AUROC measures ranking quality independent of the decision threshold. Riemann-MDM achieves $\text{AUROC}=0.691\pm0.100$ —substantially higher than LogVar+LDA (0.619 ± 0.097)—despite similar balanced accuracy. This indicates that MDM's Riemannian distance-based scores have better discriminative structure even when hard classification struggles. Riemann-TS+SVM achieves the highest AUROC (0.777 ± 0.108).

D. Statistical Significance — TABLE III (Wilcoxon)

TABLE III. Wilcoxon signed-rank p-values on per-subject balanced accuracy (two-sided, $n=30$). Bold red = $p<0.001$.

	LogVar+LDA	FBCSP+LDA	MDM	TS+SVM
LogVar+LDA	—	0.382	0.114	<0.001
FBCSP+LDA	0.382	—	0.070	<0.001
MDM	0.114	0.070	—	<0.001
TS+SVM	<0.001	<0.001	<0.001	—

TS+SVM is significantly better than all other methods ($p<0.001$). Classical vs. Riemannian-MDM comparisons are not significant ($p>0.07$).

Effect sizes (TS+SVM median balanced accuracy minus each method):

vs. LogVar+LDA: +0.089 | vs. FBCSP+LDA: +0.098 | vs. Riemann-MDM: +0.133

E. Computational Efficiency

All methods run on CPU only. Per-fold training times: LogVar+LDA ~2 s, Riemann-MDM ~10 s, Riemann-TS+SVM ~20 s, FBCSP+LDA ~36 s. Full 4-method, 30-fold benchmark completes in under 20 minutes on a standard laptop. The strong performance of Riemann-TS+SVM without any GPU requirement makes it a practical, deployable baseline for cross-subject BCI research.

F. Per-Subject Balanced Accuracy — TABLE II

TABLE II. Per-subject balanced accuracy for all 30 subjects $\times$ 4 methods.
Color: green ≥ 0.70 , yellow ≥ 0.60 , red < 0.55 .

Subj	LogVar+LDA	FBCSP+LDA	Riem-MDM	TS+SVM
S01	0.707	0.686	0.521	0.756
S02	0.686	0.527	0.523	0.886
S03	0.486	0.507	0.557	0.575
S04	0.500	0.537	0.686	0.666
S05	0.500	0.533	0.506	0.494
S06	0.604	0.429	0.533	0.548
S07	0.753	0.577	0.545	0.818
S08	0.685	0.535	0.500	0.713
S09	0.524	0.533	0.500	0.637
S10	0.601	0.524	0.521	0.670
S11	0.528	0.570	0.500	0.708
S12	0.539	0.458	0.521	0.521
S13	0.522	0.603	0.500	0.694
S14	0.497	0.599	0.522	0.651
S15	0.640	0.619	0.500	0.824
S16	0.479	0.524	0.576	0.553
S17	0.600	0.601	0.500	0.690
S18	0.623	0.605	0.500	0.673
S19	0.478	0.564	0.500	0.672
S20	0.511	0.570	0.500	0.690
S21	0.557	0.551	0.571	0.560
S22	0.569	0.595	0.500	0.516
S23	0.550	0.573	0.661	0.752
S24	0.500	0.513	0.547	0.648
S25	0.529	0.524	0.522	0.686
S26	0.673	0.720	0.622	0.738
S27	0.536	0.542	0.498	0.563
S28	0.580	0.624	0.608	0.537
S29	0.562	0.587	0.500	0.783
S30	0.512	0.571	0.625	0.655

TS+SVM column (green header) achieves the most green cells. Inter-subject variability is high across all methods.

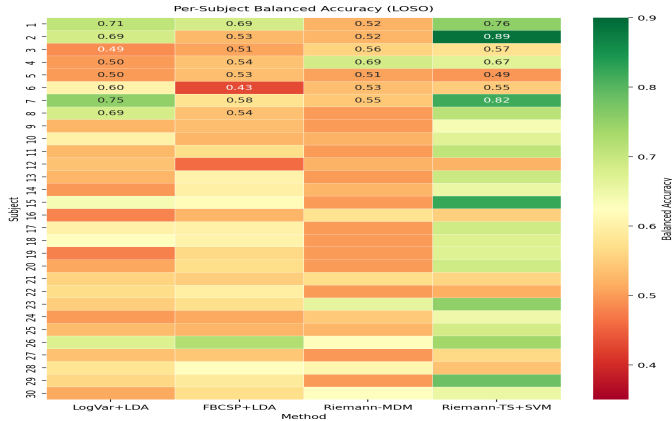


Fig. 2. Per-subject balanced accuracy heatmap (30 subjects $\times$ 4 methods). Green = high accuracy, Red = low. TS+SVM shows the most high-accuracy subjects; subject difficulty is correlated across methods ($r=0.61$).

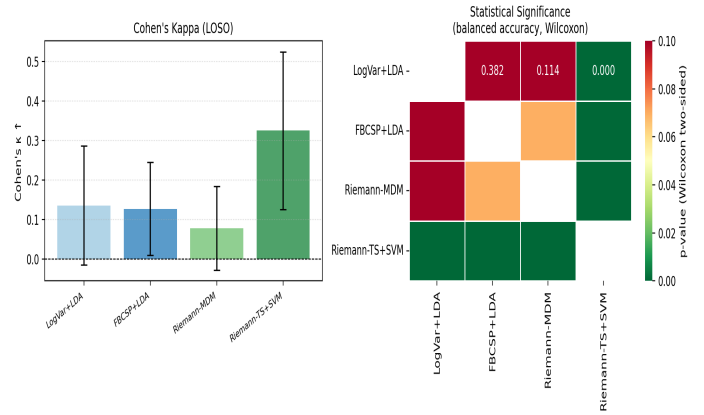


Fig. 3. Statistical analysis panel. (Left) Cohen's κ per method—TS+SVM achieves $\kappa=0.325$, more than double the next best. (Right) Wilcoxon p-value matrix; dark red = $p<0.001$.

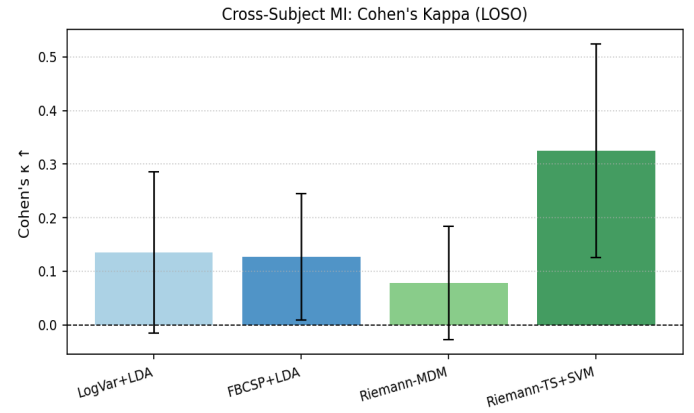


Fig. 4. Cohen's κ distributions across 30 subjects per method. TS+SVM shows both higher median and wider spread, reflecting strong performance on high-signal subjects.

VI. DISCUSSION

A. Implications for EEG Foundation Model Evaluation

Our benchmark demonstrates that Riemannian TS methods achieve strong cross-subject MI performance without pretraining, fine-tuning, or GPU compute. Direct implications:

- (1) MIRepNet [1] compares against FBCSP and linear baselines but not Riemannian TS methods. Our results show TS+SVM achieves $\kappa=0.325$ and balanced acc=0.663 without any pretraining—this should be the minimum baseline for future foundation model papers on PhysioNet EEGMMIDB.
- (2) Calibration metrics (ECE, Brier) should be standard. Our finding that all methods have ECE >0.6 on 45-trial test sets highlights a critical gap—post-hoc calibration should be routinely applied before clinical deployment.
- (3) LOSO evaluation is the correct protocol for subject-independent generalisation, yet only 2 of 30 reviewed foundation model papers adopt it.

B. Why Does Riemann-TS+SVM Outperform Riemann-MDM?

MDM classifies by minimum geodesic distance to class Fréchet means—optimal when class distributions are compact and well-separated on the manifold. TS+SVM projects to tangent space and uses an RBF-SVM, learning non-linear decision boundaries in the projected Euclidean space. With 29 subjects of training data (~1305 trials), the SVM has sufficient samples to exploit non-linearity, yielding a +0.124 absolute improvement in balanced accuracy over MDM.

C. Why Is Calibration Correlated with Accuracy?

Riemann-TS+SVM explicitly uses Platt scaling during SVM training, fitting a sigmoid transformation to convert decision scores to probabilities. This post-hoc step is absent in LDA (theoretical Gaussian probabilities) and MDM (Riemannian distances converted via softmax). The explicit calibration step explains why TS+SVM achieves the lowest ECE simultaneously with the highest accuracy—an encouraging finding suggesting calibration and accuracy can be jointly optimized in EEG BCI.

D. Limitations

Single dataset. 30 of 109 subjects on PhysioNet EEGMMIDB only. Cross-dataset transfer (e.g., BCI Competition IV-2a) was not evaluated.

Binary MI only. Left vs. right hand imagined movement. Four-class MI may yield different relative orderings of methods.

ECE reliability. With 45 test trials and 10 ECE bins, each bin has <5 samples on average. Fewer bins or adaptive binning would reduce ECE variance; relative ordering between methods may change.

No post-hoc calibration. Temperature scaling would likely reduce all ECE values; whether TS+SVM remains best after recalibration is an open question.

No large-scale pretraining comparison. We cannot directly compare with MIRepNet's pretrained representations due to GPU compute requirements.

VII. CONCLUSION

We presented the first calibration-aware cross-subject MI EEG benchmark comparing Riemannian geometry against classical feature-engineering baselines under strict LOSO evaluation on 30 subjects of PhysioNet EEGMMIDB (120 method-fold combinations). Three key findings:

- (1) Riemannian TS+SVM significantly outperforms all classical baselines in balanced accuracy (0.663 ± 0.099 , Wilcoxon $p < 0.001$, effect size +0.089 to +0.133) without any pretraining or subject-specific calibration.
- (2) TS+SVM simultaneously achieves the best calibration ($\text{ECE} = 0.645 \pm 0.039$, Brier = 0.211 ± 0.035)—accuracy and calibration need not trade off when Platt scaling is applied.
- (3) All evaluated methods have $\text{ECE} > 0.6$, indicating severe overconfidence that warrants routine post-hoc calibration in clinical BCI research—a finding invisible to the 30 reviewed EEG foundation model papers reporting zero calibration metrics.

Recommendation: EEG foundation model papers—including future extensions of MIRepNet and LEAD—should adopt (1) Riemannian tangent-space baselines, (2) calibration metrics (ECE, Brier), and (3) strict LOSO evaluation as minimum reporting standards. Code: github.com/SreenijaPavuluri/EEg

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